

Assignment - 11

Section - A

1. Complete the statements

a) rank of null matrix is 0.

b) A system is said to be consistent if it has at least one set of soln.

c) The sum of Eigen values of a matrix is equal to the trace of the matrix.

d) If 1, -2, 3 are the Eigen values of A, then the Eigen values of $3A$ are 3, -6, 9.

2. Show that the vector $(3, 1, 5)$ and $(1, 2, 1)$ are linearly independent.

Set linear combination to zero:

$$C_1(3, 1, 5) + C_2(1, 2, 1) = (0, 0, 0)$$

$$3C_1 + C_2 = 0$$

$$C_1 + 2C_2 = 0 \Rightarrow C_1 = -2C_2$$

Substitute C_1

$$3(-2C_2) + C_2 = 0$$

$$-5C_2 = 0$$

$$C_2 = 0$$

Since $C_2 = 0$, $C_1 = 0$ the vectors are linearly independent.

3 for which value of x will be the matrix
becomes singular.
for a matrix to be singular

$$\begin{bmatrix} 8 & 2 & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{bmatrix}$$

$$|A| = 0$$

$$A = \begin{bmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{bmatrix}$$

$$|A| = 8(0-12) - x(0-24) + 0 = 0$$

$$= -96 + 24x = 0$$

$$= 24x = 96$$

$$x = 4$$

4 Obtain the rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$
using Echelon form.

$$R_2 \rightarrow R_2 + 2R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 3 & 3 & -3 \\ 0 & -2 & -2 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2/3$$

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -2 & -2 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2 \text{ and } R_4 \rightarrow R_4 - R_2$$

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank = 2 ; no. of non-zero rows.

Ex Obtain the rank of matrix $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & 2 & 7 \end{bmatrix}$ using normal form.

$$R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 3 & -10 \\ 0 & 2 & -3 & 10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 3 & -10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

rank = 2 ; no. of non-zero rows.

Section - B

1 Using Elementary row transformations, find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & 1 \\ 3 & -2 & 1 \end{bmatrix}$

$$|A| = 1(1 \cdot 1 - 1 \cdot (-2)) - 2(-1 \cdot 1 - 1 \cdot 3) + 2((-1) \cdot (-2) - 1 \cdot 3)$$

$$= 1(1+2) - 2(-1-3) + 2(2-3) = 3+8-2 = \underline{\underline{9}}$$

So, $|A| = 9 \neq 0$ inverse exists.

Cofactor

$$C = \begin{bmatrix} 3 & 4 & -1 \\ -6 & -5 & 8 \\ 0 & -3 & 3 \end{bmatrix}$$

Adj (transpose of Cofactor)

$$\text{adj}(A) = \begin{bmatrix} 3 & -6 & 0 \\ 4 & -5 & -3 \\ -1 & 8 & 3 \end{bmatrix}$$

Inverse

$$A^{-1} = \frac{1}{9} \begin{bmatrix} 3 & -6 & 0 \\ 4 & -5 & -3 \\ -1 & 8 & 3 \end{bmatrix}$$

Discuss the consistency of the system and if consistent, solve the Eqn:

$$6x + y + z = -4 \quad \text{--- (1)}$$

$$2x - 3y - z = 0 \quad \text{--- (2)}$$

$$-x - 7y - 2z = 7 \quad \text{--- (3)}$$

$$2x - 3y - z = 0 \Rightarrow z = \underline{2x - 3y}$$

Substitute
in Eq (1)

$$6x + y + (2x - 3y) = -4$$

$$8x - 2y = -4$$

$$4x - y = -2$$

$$\underline{y = 4x + 2}$$

Substitute into z

$$z = 2x - 3(4x + 2) = 2x - 12x - 6 = -10x - 6$$

Put in (3)rd Eqn

$$-x - 7y - 2z = 7$$

Substitute x and z

$$-x - 7(4x + 2) - 2(-10x - 6) = 7$$

$$-x - 28x - 14 + 20x + 12 = 7$$

$$-9x - 2 = 7 \Rightarrow -9x = 9 \quad \boxed{x = -1}$$

$$y = 4(-1) + 2 = \underline{\underline{-2}}$$

$$z = -10(-1) - 6 = \underline{\underline{4}}$$

$$x = \underline{\underline{-1}}$$

3 For what values of K , the system of simultaneous

$$x + y + z = 1 \quad \text{--- (1)}$$

$$2x + y + 4z = K \quad \text{--- (2)}$$

$$4x + y + 10z = K^2 \quad \text{--- (3)}$$

have soln and solve them completely in each case.

Subtract (2) - (1)

$$x + 3z = K - 1 \quad \text{--- (4)}$$

(3) - (2)

$$2x + 6z = K^2 - K \Rightarrow x + 3z = \frac{K^2 - K}{2} \quad \text{--- (5)}$$

For system to have a soln.

$$(4) = (5)$$

$$K - 1 = \frac{K^2 - K}{2}$$

$$2K - 2 = K^2 - K \Rightarrow K^2 - 3K + 2 = 0$$

$$= (K - 1)(K - 2) = 0$$

$$\underline{\underline{K = 1, 2}}$$

Section-C

Find the Eigen Values and Eigen vectors of the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$$|A - \lambda I| = -\lambda^3 + 3\lambda + 2 = 0$$

$$= (\lambda - 2)(\lambda + 1)^2 = 0$$

$$\lambda = 2, -1, -1$$

for $\lambda = 2$

~~$$X = [1 \ 1 \ 1]$$~~

$$(A - 2I)x = 0$$

~~$$x = y = z$$~~

~~$$\text{Eigen vector} = (1, 1, 1)$$~~

for $\lambda = -1$

$$x + y + z = 0$$

$$(1, -1, 0), (1, 0, -1)$$

2. Find the Eigen value and Eigen Vectors of the matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$

$$|A - \lambda I| = 0$$

$$(\lambda - 1)^2(\lambda - 2) = 0$$

$$\underline{\lambda = 1, 1, 2}$$

for $\lambda = 2$

$$(2, 1, -1)$$

for $\lambda = 1$

$$(1, 0, -1), (0, 1, -1)$$

3. State Cayley Hamilton theorem and verify it for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1} .

$$\boxed{f(A) = 0}$$

$$|A - \lambda I| = 0$$

$$= (\lambda - 1)^2 (\lambda - 4) = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

$$A^3 - 6A^2 + 9A - 4I = 0$$

Multiply both side by A^{-1}

$$A^2 - 6A + 9I - 4A^{-1} = 0$$

$$A^{-1} = \frac{1}{4} (A^2 - 6A + 9I)$$

$$A^2 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

Substitute

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

~~Ans~~
5/8/26

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